

OCCURRENCE AND IMPORTANCE OF CUCURBITS DOWNY MILDEW DISEASE IN EGYPT

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Abstract

Downy mildew of cucurbits, caused by *Pseudoperonospora cubensis* (Berk. & Curt.) Rostow, is one of the most important economic diseases which attack cucurbits causing great losses in the yield under open field. The highest disease severity was recorded in Nily season, while the lowest was in Summer season. Percentage of disease severity at Kalubia governorate was less than those at Sharkia and Ismailia in Nily season. The highest percentage of disease severity was obtained when the plants were inoculated with 7.5×10^3 sporangia/ml, while 2.5×10^3 sporangi/ml gave the lowest percentage. High disease severity was reported on cucumber, snakemelon and cantaloupe, while the lowest disease severity was on watermelon, loofah and squash plants.

INTRODUCTION

Cucurbits downy mildew caused by *Pseudoperonospora cubensis* (Burk. & Curt.) Rostow was found in Cuba during 1868 and 20 years later in Japan, then the disease has been reported from different areas of the world. Cucumber and muskmelon are the most seriously affected host plants in the United States. Other hosts are watermelon, squash, pumpkin, gourd, wild cucumber, star cucumber and other species of the family Cucurbitaceae. In Egypt, Jones (1935) found that, watermelon was liable to infection with downy mildew.

Mahrous et al. (1985) reported that the downy mildew of cucurbits incited by

Pseudoperonospora cubensis has become a very serious problem in cucumber plantations during fall seasons. It prevails under Barrage, Nubaria and Kassassin condition, while no disease symptoms were observed under Sids and Giza conditions.

Tasai (1987) found that lesion formation and sporulation were observed 4-5 days and 5-6 days, respectively after inoculation.

Ullasa and Amin (1988) noticed that day temperature of 25-30°C night temperature of 15-21°C and RH above 75% favoured the infection of *Luffa acutangula* with *P. cubensis*.

El-Korachy (1991) reported that the highest number of sporangia of *P. cubensis* was found 7 days after inoculation at 25°C. He added that the optimum temperature for spore germination was found to be 20 °C, while no spore germination was observed at 0,35 or 40°C. The highest infection was observed at 95% relative humidity. The same author added that *Cucumis melo* var. *reticulatus* (Cantaloupe), *Cucumis sativus* (Cucumber) and *Cucumis melo* var. *inodorus* were the most susceptible hosts, while *Citrullus lanatus* (Watermelon), *Cucurbita pepo* (Squash) and *Cucurbita* sp. (Pumpkin) showed moderate susceptibility.

The present investigation aimed to carry out field survey of cucurbits downy mildew disease and its relationship to the environmental conditions in certain governorates, Kalubia, Ismailia and Sharkia, as well as, its host range.

MATERIALS AND METHODS

1- Field survey of cucurbits downy mildew and its relationship to the environmental conditions :

A quantitative survey of downy mildew disease on cucumber plants (*Cucumis sativus* L.) cultivar Biet Alpha, was carried out during 1991 and 1992 growing seasons. This survey was conducted at three fields in three governorates namely, Ismailia, Kalubia and Sharkia. Temperature, winds, rains, relative humidity (RH), dew point and saturated point in both Ismailia and Kalubia governorates during 1991 and 1992 seasons, were obtained from Agrometeorological Research Unit, soil and Water Research Institute, Agricultural Research Center, Giza (Table 1). Disease severity was estimated according to Horsfall and Barratt (1945).

Table 1. Temperature, winds, rains, relative humidity (R.H.), dew point and saturation point at Ismailia and Kalubia governorates during certain months of 1991 and 1992 season.

Governorates	Months	1991						1992									
		Temperature			Winds m/mr	Rains	R.H	Dew point	Saturation point	Temperature			Winds m/mr	Rains	R.H	Dew point	Saturation point
		Max.	Min	Mean day.						Max.	Min	Mean day.					
Ismailia	April	28.2	12.2	20.9	23	0.4	47.0	8.6	13.6	25.9	13.7	19.7	2.5	-	52.0	9.2	12.2
	May	31.1	17.4	23.7	24	-	44.0	10.0	17.6	30.3	17.2	23.3	2.4	-	49.3	11.0	16.1
	June	34.4	20.4	26.7	21	-	52.0	15.3	17.0	34.9	21.6	27.8	2.4	-	51.7	16.0	20.4
	October	30.9	18.7	24.0	15	-	65.5	17.1	10.5	30.7	17.6	23.7	1.4	-	67.0	16.7	11.1
	November	25.3	12.3	19.0	29	12.5	67.7	12.9	7.3	25.7	14.5	19.4	1.7	-	61.7	11.4	9.5
	December	23.6	10.4	19.9	31	2.9	59.0	6.0	6.6	20.6	10.1	14.8	1.9	2.2	68.5	9.2	5.9
Kalubia	April	28.3	12.3	19.5	19	2.1	57.3	1.12	10.1	25.7	9.9	17.5	3.2	-	62.3	10.03	8.1
	May	30.7	15.1	22.9	23	1.0	49.7	12.2	14.6	27.7	14.0	21.8	2.2	-	53.3	13.03	11.9
	June	33.9	18.6	26.1	22	-	50.0	15.2	16.6	31.8	18.5	26.6	1.4	-	53.7	16.03	16.8
	October	30.1	16.0	22.4	13	-	65.3	16.2	9.0	30.6	16.6	22.7	1.3	-	68.3	16.2	8.6
	November	24.2	11.5	17.4	13	1.6	73.7	12.6	5.4	34.3	13.6	17.7	1.1	-	71.3	10.6	5.9
	December	17.7	8.8	12.9	14	3.4	64.7	6.0	5.5	19.8	7.7	13.3	1.6	8.6	72.0	8.2	4.3

2- Pathogenicity tests :

Plastic pots, 25 cm in diameter, were used in this study. Pots were sterilized by immersing in 5% formalin solution, then filled with (1:1) sandy-clay soil, which was previously autoclaved at 121°C for 1 hour. Surface sterilized seeds of cucumber (Biet Alpha), watermelon (Giza 1) and squash (Escandarani) were planted in the previously autoclaved soil at the rate of 5 seeds per pot.

Plants, 3 weeks after sowing, were inoculated with *Pseudoperonospora cubensis* sporangial suspension, 5×10^3 sporangia/ml (Thomas *et al.*, 1987). Four replicates were used for each particular treatment. The experiments were repeated three times and the results were presented as the means of 20 leaves obtained from 20 different plants. Lesion colour was recorded 3,5,7 and 10 days after inoculation using the modified colour index (Cohen and Rotem 1969 and Thomas 1978). The shape, angular or circular, and sporulation density were recorded, while lesion size (mm) and disease severity were measured 10 days after inoculation (Reuveni 1983).

3- Inoculum potential :

Three different inoculum concentrations of *P. cubensis* were prepared from infected cucumber plants, (2.5×10^3 and 7.5×10^3 sporangia / ml), to study their effects on the incidence of downy mildew disease on cucumber (Biet Alpha), watermelon (Giza 1) and squash (Escandarani). The plants were sprayed with the sporangial suspension after three weeks from sowing. Lesion development and disease severity were recorded as mentioned before.

4- Host range :

The response of different hosts to infection with *P. cubensis* was studied under greenhouse conditions. Five species of the family Cucurbitaceae were tested in this study, these were as follows :

1. Cucumber (*Cucumis sativus* L.) Cultivar Biet Alpha .
2. Melon (*Cucumis Melo* L.) Cultivar Honey dew .
3. Watermelon (*Citrullus lanatus* L.) Cultivar Giza 1 .
4. Squash (*Cucumis sativus* L.) Cultivar Escandarani .
5. Loofah (*Luffa aegyptica* L.) Cultivar Egyptian loofah .

Five surface sterilized seeds of each cultivar were sown in sterilized plastic pots, 25 cm in diameter. Seedlings of three weeks old were inoculated with the

spore suspension of *P.cubensis* (5×10^3 sporangia/ml.). Disease severity was recorded as previously mentioned.

RESULTS AND DISCUSSION

1- Survey of cucurbits downy mildew in the fields of three governorates :

This survey was conducted in three governorates i.e. Ismailia, Kalubia and Sharkia on cucumber plants grown under field conditions, during two successive seasons 1991 and 1992. Data in Table (2) indicated that disease severity was higher in Ismailia than in Sharkia and Kalubia. The disease was also higher in season 1991 than in 1992. However, the infection was high in the Nily season (November and December), while it was low in the Summer season (April and May). The obtained data also reveal that there were no significant differences in disease severity among the three governorates during the Nily season, as well as in the interaction between the governorats and the growing seasons. In addition, the interaction between Summer season and Nily season in 1992 was also non significant. This increase in disease severity may be due to the compatibility of meteorological conditions with the disease incidence, as presented in Table (1) which indicated that there was a slight increase in relative humidity (R.H.) during October, November and December of season 1992 as compared with season 1991, which may explain the increase in the disease severity in the Nily plantation comparing with the Summer season. These results are in agreement with those obtained by Dercole (1975), Mahrishi and Siradhana (1988), Ullasa and Amin (1988), Huang et al. (1989) and Tsai and Hsu (1989).

2- Pathogenicity tests :

Three plant species belonging to the family Cucurbitaceae i.e *Cucumis sativus* (Cucumber), *Citrullus* sp. (Watermelon) and *Cucurbita* sp. (Squash) were tested for etiological studies. Data presented in Table (3) indicated that the pathogen significantly attacked cucumber plants causing 32.15% disease severity, followed by watermelon (10.03%), while the lowest disease severity was recorded on squash plants (4.75%). The percentage of disease severity on the three plant species showed significant differences. The production of sporangia was abundant on cucumber but few on squash at both sporangial concentrations of 5×10^3 and 7.5×10^3 sporangia / ml. However, the lesions appeared only on cucumber, 3 days after inoculation and

Table 3 . Pathogenicity test of *Pseudoperonospora cubensis*, the causal organism of cucurbits downy mildew on three genera from the family Cucurbitaceae, with concentration of 5×10^3 sporangia/ml.

Genera and Cultivar.	Lesion colour days after inoculation				Lesion shape	Lesion size (mm)	Sporangia production	Disease severity %
	3	5	7	10				
Cucumber (Biet Alpha)	Y	Y	B	B	Ang.	9-9.5	Abundant	32.15
Water melon (Giza 1)	-	YL	YL	YL	Ang.	4.5-4.8	Moderate	10.03
Squash (Escandarani)	-	-	B	B	Circ	1-1.5	Few	4.75

L.S.D at 5% 4.41 1% 6.67
 for disease severity

Ang = Angular.
 Circ = Circular.
 YL = Yellowish.
 Y = Yellow.
 B = Brown.

later on both watermelon and squash, 5 and 7 days after inoculation, respectively. Lesions were angular in shape and large in size on cucumber and watermelon, while they were circular in shape and smaller on squash plants. In this respect, the difference between different plant species (family Cucurbitaceae) in their response to infection could be attributed to differences in the physiological and morphological characters of these plants.

3- Inoculum potential :

Results in Table (4) show that, the highest percentage of disease severity (30.77%) was obtained when plants were inoculated with 7.5×10^3 sporangia / ml, followed by 5×10^3 (15.68%), while 2.5×10^3 caused the lowest percentage of disease severity (11.20%). Thus, the differences between the effects of the three concentrations of inocula were highly significant. The first appearance of the lesions on cucumber and watermelon plants was 3 days after inoculation with the high concentrations of inocula, 5×10^3 and 7.5×10^3 sporangia/ml, while lesions appeared 5 days after inoculation with 2.5×10^3 sporangia/ml. On the other hand, lesions appeared on squash plants 7 days after inoculation with each of 2.5×10^3 and 5×10^3 sporangia/ml and 5 days after inoculation with 7.5×10^3 sporangia/ml. The highest percentage of the disease severity was observed in case of cucumber plants under all investigated inoculum concentrations. It is also clear that, the differences between the tested plant species were significant. The early appearance of disease symptoms may be due to the high concentration of the sporangia. These results were in complete agreement with those obtained by Duvdevani *et al* (1946) and Cohen (1977).

4- Host range :

To determine the host range of *P.cubensis*, five different species of cucurbits were investigated. The lesions development and disease severity are presented in Table (5). Data indicated that, the first symptoms appeared on cucumber and snake-melon three days after inoculation, and after five days on watermelon and cantaloupe, while it was delayed till 7 days on squash and loofah. Data also indicated that the lesion shapes were angular on all tested cultivars, except squash plants where it was circular. On the other hand, the lesion size showed the largest diameter on cucumber, snake-melon and cantaloupe (9-9.5, 8-8.5 and 6-6.3 mm), respectively. The smallest lesions were on squash, loofah and watermelon (1-1.5, 2.5-3.8 and 4.3-5 mm, respectively). The presented data also showed that the cucumber, cantal-

Table 4 . Effect of different inoculum concentrations of *Pseudoperonospora cubensis* on the incidence of downy mildew disease on three plant species from the family Cucurbitaceae.

Concentration (C)		Lesion colour days after inoculation				Lesion shape	Lesion size (mm)	Sporangia production	Disease severity %
	Genera (G)	3	5	7	10	After 10 days from inoculation			
	Cucumber (Biet Alpha)	-	YL	YL	B	Ang.	4-65	Moderate	20.50
	Water melon (Giza 1)	-	G	G	YL	Ang.	2-25	Few	8.30
	Squash (Escandarani)	-	-	G	YL	Circ	1-13	Few	4.80
Mean									11.20
2.5 x 10 ³ sporangial/ mL.	Cucumber (Biet Alpha)	Y	Y	B	B	Ang.	4-9.5	Abundant	32.20
	Water melon (Giza 1)	YL	Y	Y	Y	Circ	4.5-4.8	Few	10.30
	Squash (Escandarani)	-	-	Y	B	Circ	1-1.5	Few	4.80
	Mean	15.68							
2.5 x 10 ³ sporangial/ mL.	Cucumber (Biet Alpha)	YL	Y	B	B	Ang.	5.8-6.5	Abundant	61.40
	Water melon (Giza 1)	YL	YL	B	B	Circ	2.8-3	Moderate	22.60
	Squash (Escandarani)	-	YL	Y	B	Circ	1.3-1.8	Few	8.30
	Mean	30.77							

L.S.D at 5% 1% Circ. = Circular G = Green
 for disease severity YL=Yellowish
 Genera (G) = 1.44 2.18 Ang. = Angular Y = Yellow
 Concentration (C) = 1.57 2.15 B = Brown
 (G x C) = 2.71 3.72

7.5 x 10³ sporangial/mL.5 x 10³ sporangial/mL.2.5 x 10³ sporangial/mL.

loupe and snakemelon cultivars were abundant in sporangial production, while watermelon, squash and loofah showed less numbers.

It is clear from results in Table (5) that the highest disease severity had been recorded on cucumber being 35.4% and snakemelon (30.5%) followed by cantaloupe (19.2%), while the lowest disease severity was recorded on squash followed by loofah (5.6 and 9.3%), respectively. Differences between different hosts of family Cucurbitaceae in their reactions to downy mildew could be attributed to the physicochemical processes of the diseased plants such as photosynthesis, respiration and nitrogen metabolism (Mehrotra 1983). Resistance or susceptibility may be conditioned by a number of internal or external factors which operate to reduce or increase the chance and degree of infection. Several kinds of resistance may be physiological or chemical and may depend upon the lack of a nutrient or a substance that the parasite needs for development or it may be due to a toxic or repellent substance that is either preformed or formed only in response to infection. Similar results were obtained by Mahrous *et al.* (1985) and El-Korachy (1991). The results indicated that, the most important economic hosts for *P. cubensis* included cucumber, melon, snakemelon, watermelon, squash and loofah, as also suggested by several workers (Palti and Cohen 1978, Mahrishi and Siradhana 1988 and El-korachy 1991).

Generally, it could be concluded that the highest disease severity was observed in November and December (Nily season) at the three governorates in the two successive seasons of 1991 and 1992. The percentage of disease severity at Kalubia governorate was the least when compared with the percentages recorded at Sharkia and Ismailia, during the Nily season. The percentage of disease severity increasing the inoculum level. The genus *Cucumis* sp. was more susceptible to infection than other genera under artificial infection conditions in the greenhouse. The lesion was angular in shape on all the tested cultivars, while it was circular on squash plants. In addition, sporangial production was abundant on cucumber, cantaloupe and snakemelon, *Cucumis* sp., while it was few on watermelon, loofah and squash.

Table 5 . Host range of *P. cubitriss* on different genera of Cucurbitaceae under artificial inoculation in the greenhouse, with 5 x 10³ sporangia / ml.

Genera and Cultivar.	Lesion colour days after inoculation				Lesion shape	Lesion size (mm)	Sporangia production	Disease severity %
	3	5	7	10				
Cucumber (Biet Alpha)	Y	Y	B	B	Ang.	9-9.5	Abundant	35.4
Water melon (Giza 1)	-	YL	YL	YL	Ang.	4.3-5	Few	16.02
Squash (Escandarani)	-	-	B	B	Circ.	1-1.5	Few	5.6
Cantaloupe (Honey dew)	-	Y	B	B	Ang.	6-6.3	Abundant	19.2
Egyptian lootah	-	-	B	B	Ang.	2.5-3.8	Few	9.3
Snakemelon	Y	Y	B	B	Ang.	8-8.5	Abundant	30.5

L.S.D at 5% 1%
for disease severity 3.49 4.90

Ang = Angular.
Circ = Circular.
YL = Yellowish.
Y = Yellow.
B = Brown.

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أهمية وحدوث مرض البياض الزغبى فى القرعيات فى ج.م.ع

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يعتبر مرض البياض الزغبى فى القرعيات المتسبب عن الفطر سيدروبيرونوسورا كيوبنسيس واحدا من أهم الأمراض النباتية الاقتصادية التي تهاجم القرعيات وتسبب خسارة كبيرة فى المحصول الثمرى تحت ظروف الحقل المفتوح. أوضحت النتائج أن أعلى نسبة إصابة بالمرض سجلت فى العروة النيلية (نوفمبر وديسمبر) فى كل المحافظات المختبرة فى موسمى ١٩٩١، ١٩٩٢ بينما أقل نسبة إصابة كانت فى العروة الصيفى (أبريل ومايو). وكانت محافظة القليوبية أقل إصابة عن محافظتى الاسماعيلية والشرقية خاصة فى العروة النيلية . وجد أن أعلى شدة إصابة بالمرض عند عدوى النباتات بتركيز ٧,٥ x ١٠^٢ اسبورانجيا / مل ، وذلك تحت ظروف العدوى الصناعية فى الصوبة، كما وجد أن الأنواع التابعة لجنس *Cucumis* sp. (الخيار - القناء - الكانتلوب) أكثر إصابة بالمرض عن الأنواع التابعة للأجناس الأخرى (البطيخ) *Citrullus* s و (الكوسة) *Cucurbita* sp. و (اللوب) *Luffa* sp.